

HW 9

Review Questions and Exercises

2. No, a structure declaration just defines a template for a new data type; it doesn't allocate memory for a variable.
3. An array stores multiple values of same data type, whereas a structure can store multiple values of different data types.
7. Canton; Hoywood; 9478; 0
11. defined
12. structure variable name
18. Car array car[25];
19. Car array car[35] =
{
 {"Ford", "Taurus", 1997, 21000.0},
 {"Honda", "Accord", 1992, 11000.0},
 {"Lamborghini", "Countach", 1997, 200000.0},
};
30. T
34. T
46. Can't initialize array in structure declaration
47. should define a variable, not use structure name directly
50. can't left empty between 0 and 4
51. should return things

Programming Challenges

```
1  #include <iostream>
2  #include <iomanip>
3  using namespace std;
4  struct Computer
5  {
6      double speed, price;
7      int cores, ram, disk, benchmark;
8  };
9  Computer inventory[100];
10 int total = 0;
11 double getValidInput(string prompt)
12 {
13     double val;
14     do
15     {
16         cout << prompt;
17         cin >> val;
18         if (val < 0)
19             cout << "invalid input\n";
20     } while (val < 0);
21     return val;
22 }
23 void addConfig()
24 {
25     if (total >= 100)
26     {
27         cout << "inventory full\n";
28         return;
29     }
30     cout << "input speed(GHz), cores, RAM(GB), disk(GB): ";
31     cin >> inventory[total].speed >> inventory[total].cores >> inventory[total].ram >> inventory[total].disk;
32     inventory[total].benchmark = (int)getValidInput("input benchmark: ");
33     inventory[total].price = getValidInput("input price: ");
34     total++;
35 }
36 void updatePrice()
37 {
38     int id;
39     cout << "input id number: ";
40     cin >> id;
41     if (id >= 1 && id <= total)
42     {
43         inventory[id - 1].price = getValidInput("input new price: ");
44     }
45     else
46     {
47         cout << "invalid id\n";
48     }
49 }
50 void showData()
51 {
52     for (int i = 0; i < total; i++)
53     {
54         cout << "number: " << (i + 1) << endl;
55         cout << " speed: " << inventory[i].speed << " GHz" << endl;
56         cout << " core: " << inventory[i].cores << endl;
57         cout << " RAM : " << inventory[i].ram << " GB" << endl;
58         cout << " disk: " << inventory[i].disk << " GB" << endl;
59         cout << " benchmark: " << inventory[i].benchmark << endl;
60         cout << " price: $" << fixed << setprecision(2) << inventory[i].price << endl;
61     }
62 }
63 int main()
64 {
65     int choice;
66     do
67     {
68         cout << "\n1. add computer\n2. change price\n3. show all\n4. exit\nchoice: ";
69         cin >> choice;
70         switch (choice)
71         {
72             case 1:
73                 addConfig();
74                 break;
75             case 2:
76                 updatePrice();
77                 break;
78             case 3:
79                 showData();
80                 break;
81             case 4:
82                 cout << "exit...\n";
83                 break;
84             default:
85                 cout << "invalid\n";
86         }
87     } while (choice != 4);
88     return 0;
89 }
```

```
1. add computer
2. change price
3. show all
4. exit
choice: 1
input speed(GHz), cores, RAM(GB), disk(GB): 1
1
1
1
input benchmark: 1
input price: 2

1. add computer
2. change price
3. show all
4. exit
choice: 3

number: 1
 speed: 1 GHz
 core: 1
 RAM : 1 GB
 disk: 1 GB
 benchmark: 1
 price: $2.00

1. add computer
2. change price
3. show all
4. exit
```